

Systematic Literature Review Protocol

1. Document Title

Systematic Literature Review Protocol: Predictive Machine Learning Architectures, Class Imbalance Mitigation, and Explainable Analytics in Higher Education Student Dropout Prediction

2. Research questions:

- **RQ1:** What predictor variables, feature taxonomy dimensions, and algorithmic architectures dominate empirical dropout prediction literature, and how do they inform an integrated framework?
 - **RQ2:** How can generative adversarial tabular oversampling (CTGAN) and complementary model-specific and modelagnostic explainability methods be integrated into a late-fusion engine to mitigate minority class misclassification and algorithmic opacity without introducing temporal data leakage or hyperparameter optimization bias?
 - **RQ3:** What protocol and data governance requirements are necessary to evaluate the institutional transferability of a predictive learning analytics design across distinct university contexts?
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3. Review Objectives

Primary Objective

To systematically map, evaluate, and synthesize empirical research on machine learning-based student dropout prediction in higher education, establishing an evidence-based foundation for the conceptual design, algorithmic choices, data governance rules, and explainability requirements of the **Explainable Higher-Education Dropout Prediction Framework (XHDPF)**.

Specific Objectives

- **Predictor & Algorithmic Taxonomy (RQ1):** Identify and classify the primary predictor variables (Socio-demographic, Academic Performance, LMS Engagement, Financial/Scholarship), feature taxonomy dimensions, and algorithmic classification architectures (e.g., Random Forest, XGBoost, Stacked LSTM) dominating empirical dropout prediction literature.
- **Class Imbalance & Explainability Integration (RQ2):** Evaluate how class-imbalance mitigation techniques (comparing linear interpolation like SMOTE against deep

generative tabular oversampling like CTGAN) and multi-level explainable AI (XAI) methods (TreeSHAP, LIME) are integrated into predictive pipelines without introducing temporal data leakage.

- Governance & Institutional Transferability (RQ3): Synthesize systemic reporting gaps regarding administrative data governance (DAMA-DMBOK), cryptographic student privacy (SHA-256), subgroup fairness auditing (ΔEOD), Disparate Impact), and cross-institutional domain adaptation protocols.

4. PICOC Framework Operationalization

The systematic literature review protocol is formalized using the **Population, Intervention, Comparison, Outcome, and Context (PICOC)** framework:

PICOC Element	Operational Specification
Population (P)	Undergraduate students enrolled in accredited higher education institutions (universities, colleges, and tertiary institutes).
Intervention (I)	Predictive machine learning, educational data mining, and learning analytics pipelines, algorithms, or integrated frameworks deployed to predict early student attrition and course/degree dropout.
Comparison (C)	Traditional statistical baselines (e.g., logistic regression), unmitigated imbalanced datasets, or conventional predictive workflows operating without data governance, deep generative balancing, or model explainability.
Outcome (O)	Primary Outcome: Definitive binary student dropout ($y = 1$) versus active retention ($y = 0$). Secondary Outcomes: Classification performance metrics (AUC-ROC, PR-AUC, F1-Score, Recall, ECE calibration), class-imbalance mitigation efficacy, local/global feature attributions, and subgroup fairness parity.
Context (C)	Higher education institutional environments (campus-based, distance-learning, and hybrid modalities), with specific focus on Latin American tertiary education contexts and cross-institutional transferability scenarios.